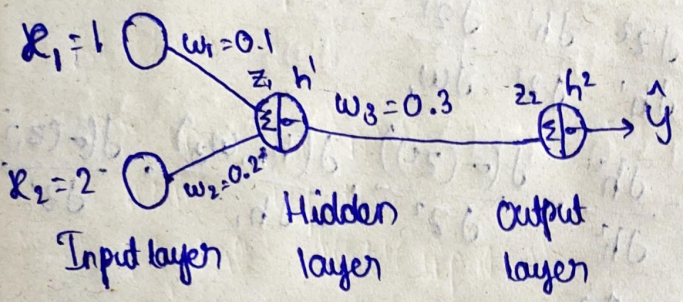
Forward & Backward Propagation

$y_{act} = 1$



* 2 inputs neurons
* 1 hidden neurons
* 1 output neurons
* NO bias terms

* Sigmoid activation function * mean Square Error loss

Given :

$x_1 = 1$ $\quad w_1 = 0.1$

$x_2 = 2$ $\quad w_2 = 0.2$ $\quad$ Learning rate $\eta = 0.1$

$y\text{-}act = 1$ $\quad w_3 = 0.3$

* Forward Propagation

$$z_1 = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow [w_1, w_2]^T \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow [w_1 x_1 + w_2 \times x_2]$$

$$= [0.1 \times 1 + 0.2 \times 2]$$

$$= [0.1 + 0.4]$$

$$z_1 = 0.5$$

$$h^1 = \frac{1}{1+e^{-0.5}} \Rightarrow 0.6225 \qquad \boxed{\sigma = \frac{1}{1+e^{-x}}}$$

$$z_2 = 0.6225 \times 0.3 \Rightarrow 0.18675$$

$$h^2 = \frac{1}{1+e^{-0.18675}} \Rightarrow 0.5466 \Rightarrow \hat{y}$$

* Loss Calculation (MSE) $= \frac{1}{m} \sum_{i=1}^{m} (y_i - \hat{y}_i)^2$

$$\text{Loss} = \frac{1}{2}(y - \hat{y})^2$$

$$= \frac{1}{2}(1 - 0.5466)^2$$

$$L = \frac{1}{2}(0.4534)^2 = 0.1028.$$

Back propagation :-

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h^2} \cdot \frac{\partial h^2}{\partial z^2} \frac{\partial z^2}{\partial h^1} \frac{\partial h^1}{\partial z^1} \cdot \frac{\partial z^1}{\partial w_1}$$

$$= \frac{\partial (y-\hat{y})^2}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h^2} = \frac{\partial h^2}{\partial h^2} \cdot \frac{\partial (\sigma(z^2))}{\partial z^2} \cdot \frac{\partial (h_1 \cdot w_3)}{\partial h_1} \cdot \frac{\partial (\sigma(z_1))}{\partial z^1}$$

- $$\frac{\partial (y-\hat{y})^2}{\partial \hat{y}} = 2(y-\hat{y}) \frac{\partial (y-\hat{y})}{\partial \hat{y}}$$

$$= 2(y-\hat{y}) \left[\frac{\partial y}{\partial \hat{y}} - \frac{\partial \hat{y}}{\partial \hat{y}} \right]$$

$$= 2(y-\hat{y}) [0-1]$$

$$= -2(y-\hat{y})$$

$$\boxed{\cancel{2(1-0.5465)} = -0.907}$$

- $$\frac{\partial \hat{y}}{\partial h^2} = \frac{\partial h^2}{\partial h^2} = 1 \qquad \boxed{\because \hat{y} = h^2}$$

- $$\frac{\partial h^2}{\partial z^2} = \frac{\partial (\sigma(z^2))}{\partial z^2} = \sigma(z^2)(1-\sigma(z^2)) \qquad \boxed{\because h^2 = \sigma(z^2)}$$

$$= \sigma(0.1866)(1-\sigma(0.1866))$$

$$= 0.546\ (1-0.546)$$

$$= 0.2479$$

- $$\frac{\partial z^2}{\partial h^1} = \frac{\partial (h_1 w_3)}{\partial h_1} = w_3 = 0.3$$

$$\frac{\partial z^2}{\partial h^1} = 3$$

- $$\frac{\partial h^1}{\partial z^1} = \frac{\partial (\sigma(z_1))}{\partial z^1}$$

$= \sigma(z_1)(1-\sigma(z_1))$

$= \sigma(0.5)(1-\sigma(0.5))$

$= 0.622(1-0.622)$

$= 0.622(0.235)$

$$\frac{dh'}{dz'} = 0.235$$

$$\frac{dz'}{dw_1} = \frac{d}{dw_1}(w_1 x_1 + w_2 x_2)$$

$$= x_1$$

$$\frac{dz'}{dw_1} = 1$$

$$\frac{dL}{dw_1} = -2(y-\hat{y}) \times (1) \times (0.2479 \times 0.3 \times 0.235 \times 1)$$

$$= 2(1-0.5465) \times 0.2479 \times 0.3 \times 0.235$$

$$= -2(0.454) \times 0.248 \times 0.3 \times 0.235$$

$$\frac{dL}{dw_1} = -0.0158$$

update weight:

$$w'_{new} = w' - \eta\left(\frac{dL}{dw'}\right) \quad [\because \eta = 0.1]$$

$$w_1 = w_1 - \eta(-0.0158)$$

$$= 0.1 - (0.1)(-0.0158)$$

$$= 0.1 + 0.00158$$

$$w_1 = 0.10158$$

$$\frac{dL}{dw_2} = \frac{dL}{d\hat{y}} \times \frac{d\hat{y}}{dh^2} \times \frac{dh^2}{dz^2} \times \frac{dz^2}{dh^1} \times \frac{dh^1}{dz_1} \times \frac{dz^1}{dw_2}$$

$$= -2(0.454) \times 1 \times 0.2479 \times 0.3 \times 0.235 \times 2$$

$$\frac{dL}{dw_2} = -0.0817$$

Update weight :-

$$w^2_{new} = w^2_{old} - \eta\left(\frac{dL}{dw_2}\right)$$

$$= 0.2 - (0.1)(-0.0317)$$

$$= 0.2 + 0.00316$$

$$w^2_{new} = 0.20316$$

$$\frac{dL}{dw_3} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dh_2} \cdot \frac{dh_2}{dz^2} \cdot \frac{dz^2}{dw_3} \qquad \frac{d(w_3 \cdot h_1)}{dw_3} = h_1$$

$$= -2(0.454) \times 1 \times 0.248 \times 0.622$$

$$\frac{dL}{dw_3} = -0.140.$$

Update weight :

$$w^2_{new} = w^3 - \eta\left(\frac{dL}{dw_3}\right)$$

$$= 0.3 - (0.1)(-0.140)$$

$$= 0.3 + 0.014$$

$$w^3_{new} = 0.314$$